

```

1  #include <stdio.h>
2  #include <stdlib.h>
3  struct node
4  {
5      int data;
6      struct node *next;
7  };
8
9  struct node *head = NULL;
10
11 void push()
12 {
13     int value;
14     struct node *newNode;
15     newNode = (struct node *)malloc(sizeof(struct node));
16     if(newNode == NULL){
17         printf("memory insufficient\n");
18         return;
19     }
20     printf("Enter the data: ");
21     scanf("%d",&value);
22     newNode ->data = value;
23     newNode ->next = NULL;
24     if(head == NULL) {
25         head = newNode;
26     }
27     else {
28         newNode->next = head;
29         head = newNode;
30     }
31     printf("%d is inserted\n",value);
32 }
33
34 void pop()
35 {
36     struct node *temp;
37
38     if(head == NULL){
39         printf("Stack is empty\n");
40         return;
41     }
42
43     temp = head;
44     printf("%d is removed\n", temp->data);
45     head = head->next;
46     free(temp);
47 }
48
49 void peek()
50 {
51     if(head == NULL){
52         printf("Stack is empty\n");
53         return;
54     }
55     printf("Top element: %d\n", head->data);
56 }
57
58 void display()
59 {
60     struct node *temp;
61
62     if(head == NULL){
63         printf("Stack is empty\n");
64         return;
65     }
66     temp = head;

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67     while(temp != NULL){
68         printf("%d -> ", temp->data);
69         temp = temp->next;
70     }
71     printf("\n");
72 }
73
74 int main()
75 {
76     int ch;
77     while(1)
78     {
79         printf("Stack Menu\n");
80         printf("1. PUSH\n");
81         printf("2. POP\n");
82         printf("3. PEEK\n");
83         printf("4. DISPLAY\n");
84         printf("5. EXIT\n");
85         printf("Enter your choice (1/2/3/4/5): ");
86         scanf("%d",&ch);
87         switch (ch)
88         {
89             case 1:
90                 push();
91                 break;
92             case 2:
93                 pop();
94                 break;
95             case 3:
96                 peek();
97                 break;
98             case 4:
99                 display();
100                break;
101             case 5:
102                 return 0;
103             default:
104                 printf("you entered wrong choice");
105         }
106     }
107     return 0;
108 }
109 }

```